

Circle field test

Print the markers

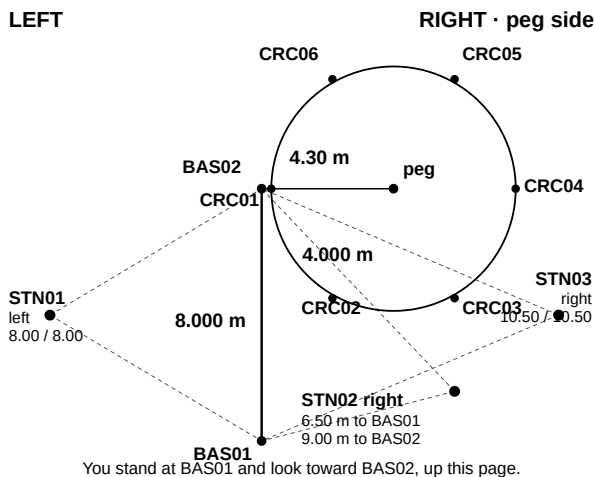
Print the sleeves the day before. **Actual size / 100%**. No fit-to-page, no scaling, no borderless. Black only, highest quality. The **100 mm** bar must measure 100 mm (± 0.5 mm), or reprint. Cut the **205 mm** line, and 5 mm off the top and the bottom. Wrap snug on a **60 mm** dowel, bands horizontal, **TOP** up. Glue the **16.5 mm** flap and press the seam flat. $R = C / 2\pi$ at three heights, to **0.1 mm**, on the cane row. Steps: [How to print the markers](#).

All sleeves. BAS01 [315](#), BAS02 [36E](#), CRC01 [459](#), CRC02 [6BA](#), CRC03 [8A3](#), CRC04 [952](#), CRC05 [DAC](#), CRC06 [EC5](#).

STN01, STN02, and STN03 are where you stand. They are not sleeved. Calibration reuses a cane. This test does not use a checkerboard.

One person. One iPhone. About an hour. Numbers are the approved field sheet. Do not change them.

Look at the plan from **BAS01** toward **BAS02** (up the page). **Left** is left. The peg is on your **right**.



Kit

- ☐ One iPhone. Rear camera, zoom **1×**, held **landscape**. Same phone for every photo, including the calibration.
- ☐ A tape that reaches **8 m** clear. STN01 uses all of that 8 m.
- ☐ A laser, or a tape that reaches **10.5 m**.
- ☐ A non-stretch line marked at **4.000 m** against the tape.
- ☐ A centre peg. Six canes for the circle. Two canes for the baseline. One cane for the calibration if you do not reuse a circle cane.
- ☐ Sleeves from **Print the markers**, glued snug on a **60 mm** dowel. Matte. Each sleeve's own R.
- ☐ A small spirit level. The phone pole (about **1 m**). This sheet and a pen.

Camera location off. Leave the app's **Offset A (mm)** and **Offset B (mm)** at 0.

Set out

1. Peg in the lawn. Line on the peg, **4.000 m** to the stick axis. Plant **CRC01** nearest the baseline.
2. Each next cane is **4.000 m** from the peg and **4.000 m** from the one before: **CRC02**, **CRC03**, **CRC04**, **CRC05**, **CRC06**. Closing chord CRC06–CRC01 within **30 mm** of 4.000 m, or walk them again.
3. Diameters CRC01–CRC04 and CRC02–CRC05, tape level, axis to axis **8.000 m**. Both within **20 mm**. Then write radius σ **3 mm**. Outside those limits: walk the sticks again. Do not write a larger σ .
4. From the peg through CRC01, **4.30 m** is **BAS02** (about **0.30 m** past CRC01). Write the distance you taped.
5. Baseline square to that line. From BAS02 back toward the peg mark **3.00 m**. From that mark swing **5.00 m**, and from BAS02 swing **4.00 m**. Through that side point, **BAS01** is **8.000 m** from BAS02.

6. Tape BAS01–BAS02 **both ways**, sleeve to sleeve, level. Level each reading, then add that end's measured R. The two axis lengths must agree within **10 mm**. The mean is the length you type. Do not average a pair that disagrees by more than 10 mm.
7. Laser the same baseline, sleeve to sleeve, same height. This is **CHK-BASE**. It is not typed into the app. Retake if the dot is more than half that sleeve's R (about **15 mm**) off the middle. Level, then add R for each sleeved end.
8. Bubble every cane (CRC01–CRC06, BAS01, BAS02, CAL). Middle of the bubble, about **1°**. A **2°** lean at 0.5 m sleeve height is **17 mm**. Write height to the middle of the paper. **0.5 m** is a good target. Tick plumb only when the bubble passed.

Photos

1×, landscape, lens glass over the middle of the pole, pole bubbled, then shoot. Stay on the pole when you turn. A second photo is the same station. Do not step. If a listed mark is cut off, re-measure. Do not add a turn.

Calibration, before any station. One plumbed cane. Front of the phone about **4.000 m** from the paper (between **3 m** and **5 m**). Tape that, level, write the raw reading. Axis length is the reading plus that sleeve's R. One still. Call it **CAL** on this sheet.

STN01 — left. 8.00 m / 8.00 m.

1. Both baseline ends, and CRC01, CRC02, CRC03, CRC04, CRC05. CRC06 does not fit.
2. Turn. Keep **BAS02**. Add CRC06.

STN02 — right. 6.50 m from BAS01, 9.00 m from BAS02. Outside the circle.

1. Both baseline ends. No circle stick between them.
2. Turn. Keep **BAS02**. Take CRC02, CRC01, CRC06, CRC03. **Do not take CRC05**. It sits behind CRC03.
3. **Do not photograph CRC04 from here.**

STN03 — right. 10.50 m / 10.50 m. Required. Two stations are not a finished test.

1. Both baseline ends, and CRC01, CRC02, CRC03.
2. Turn. Keep **BAS02**. Add CRC06, CRC05, CRC04.

Tape each station from the **pole** to the **sleeve**, level. Add R once, for the sleeve end only. Write the readings below. The app has no box for them.

In the app

This build has no probe page. Shoot in **Camera.app**, then load with **Library**.

Before you leave the house, in Camera.app: **12 MP**, **HEIF Max** off, **48 MP** off, **ProRAW** off, location off.

1. Open Garden Planting. Tap the bottom-right menu button (**Open menu**). **Tools** → **New garden**, if this phone should not keep an older garden.
2. Menu → **Sign in** → **Sign in with Microsoft**. **Version** is the file name (**1** is **/Garden Survey/garden-v1.json**, shown on the path line).
3. Menu → **Baseline**. **End A BAS01**, **End B BAS02**. **Length (m)** is the tape mean after level and +R, not the laser. **Offset A (mm) 0**, **Offset B (mm) 0**. **Measure with Tape**. **Save baseline**. Do not swap the ends.
4. **+ Point**. The **Aiming preview** opens. Tap **Cancel**. In the **+ Point** dialogue: **Name Circle**, **Geometry Circle**, then **+ Item**. **Save**. There is no box for sleeve R, sleeve height, or plumb. They stay on this sheet.

Write on this sheet

Date _____ Phone _____ Lens name _____

	CRC01	CRC02	CRC03	CRC04	CRC05	CRC06	BAS01	BAS02	CAL
Height m									
Sleeve R mm									
Plumb	☐	☐	☐	☐	☐	☐	☐	☐	☐

Baseline BAS01 → BAS02, level +R	Baseline BAS02 → BAS01, level +R	
Pulls differ by (mm). Within 10 mm	Mean in Length (m). Tape, not laser	
CHK-BASE laser, sleeve to sleeve	Diameter CRC01–CRC04, sleeves	
Long chord CRC01–CRC03 (~6.93 m)	Peg to BAS02 (aim 4.30 m)	
CAL, phone front to paper	STN01 to BAS01 / BAS02	
STN02 to BAS01 / BAS02	STN03 to BAS01 / BAS02	

Side, looking from BAS01 toward BAS02: STN01 left ☐ STN02 right ☐ STN03 right ☐

Lens over the pole, then bubble: STN01 ☐ STN02 ☐ STN03 ☐

Which photo is CAL _____ Radius σ (3 mm if the diameters and the closing chord passed) _____

Send back

OneDrive path under **Version** in **Sign in**. Version **1** is **/Garden Survey/garden-v1.json**. Also **/Garden Survey/photos/** and **/Garden Survey/photos/manifest.json**, and a photo of this filled sheet. Every **Photo** row must say **verified**. Leave the stills unedited. Do not put them in git. If sign-in is not configured, **Tools** → **Export garden.json**, and still send the sheet photo.

5. For a station, **Cancel** the preview if it opens, set **Name** to **STN01** (then **STN02**, **STN03**, **CAL**), tap **+ Item**, then the **+** on the **Point** row (title **+ Point**).
6. On **Aiming preview**, do not use the shutter. **Take photo** is aim-only and has no lens data. Tap **Library**. Pick the Camera.app still at **Actual Size**.
7. Second still at the same station: shoot it in Camera.app without stepping, then **+ Photo** on that point, then **Library** again.
8. Tap the thumbnail. The screen is **Mark BAS01–BAS02**. Tap **A** on BAS01 and **B** on BAS02, then **Confirm**. **Target**, **Suggest**, **Clear**, and **Cancel** are the other buttons. There is no CRC button. Write which sticks were in the frame on this sheet.
9. The **Photo** row must read **verified**, and the line under it **Upload verified**. If it says **failed**, tap **Retry**. If the message is **Sign in to upload.**, menu → **Sign in** → **Sign in**. A line **Not usable for focal length**. means retake at 1× in Camera.app and pick **Library** at **Actual Size** again.
10. Menu → **Sign in** → **Save**. Wait until it says the file was saved to OneDrive. The path line is the file you send.

There is no left/right control. Tick the side on this sheet. Do not type 0 or 1. Do not type the laser, the check diameter, or the check chord.